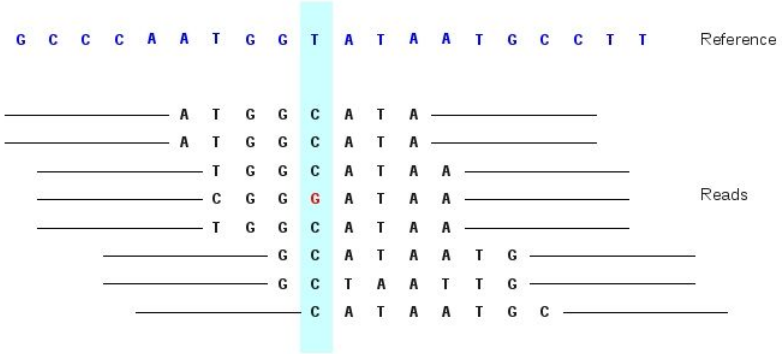
BME 105: Genetics in the Genomics Era

Lecture 5: Identifying genetic variants, part 2 (next-gen seq)



Instructor: Dr. Shelbi Russell,
Assistant Professor

Course website on canvas.ucsc.edu

Some slides modified from Angela Brooks

Roadmap: DNA sequencing and analysis for genetic variant identification

First-generation DNA sequencing (1977)
(see Lecture 4)

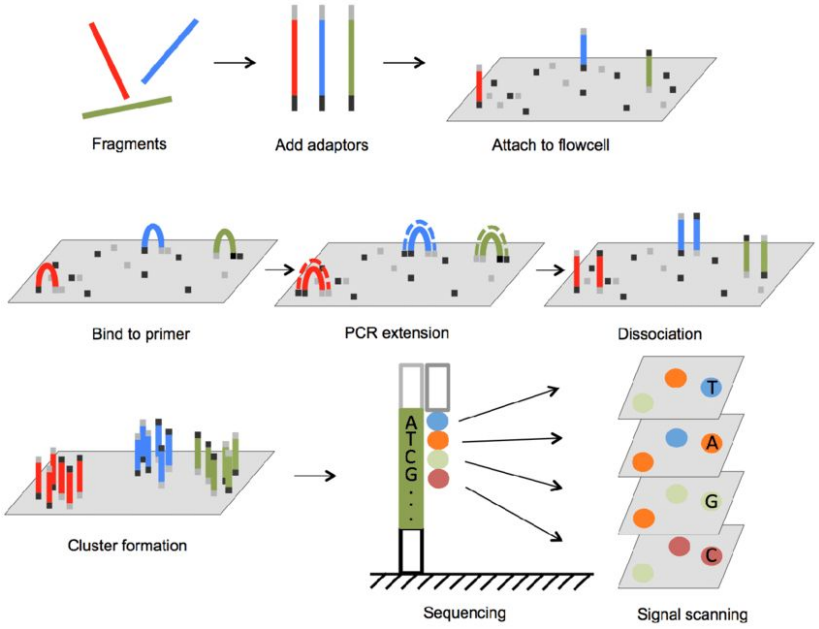
- Sanger sequencing

Second-generation DNA sequencing (~2000s)

- Massively parallel → high throughput
- One main platform until recently
  - Illumina sequencing (acquired Solexa)
  - Illumina's main competitor: MGI

Third-generation DNA sequencing (~2010s)

- Single-molecule
- Two main platforms:
  - Oxford Nanopore Technologies
  - PacBio



Illumina Sequencing method

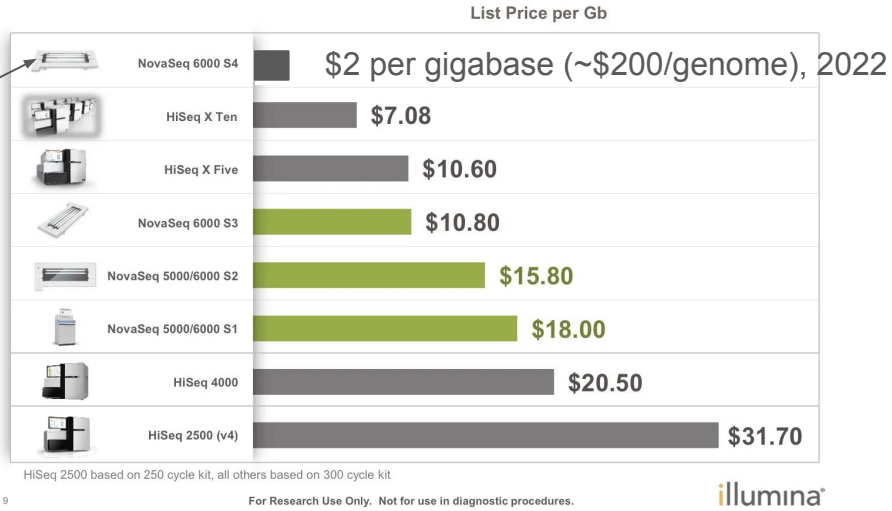
Lecture 5 Learning Objectives

- Become familiar with the types of DNA sequencing used for genome assembly (post 2000): short and long read sequencing technologies.
- Understand how Illumina "sequencing by synthesis" works.
- Understand how samples can be multiplexed and demultiplexed on an Illumina flow cell.
- Know what bacterial artificial chromosomes (BACs) are and how they are used to sequence human reference genomes
- Know the difference between a reference genome and a resequenced genome
- Understand how exome sequencing works, and how it is another use of oligo probe hybridization
- Know the basic steps and their functions in a variant calling pipeline for short-read data
- Be able to describe why long reads, many kilobases long, were necessary to complete the human genome

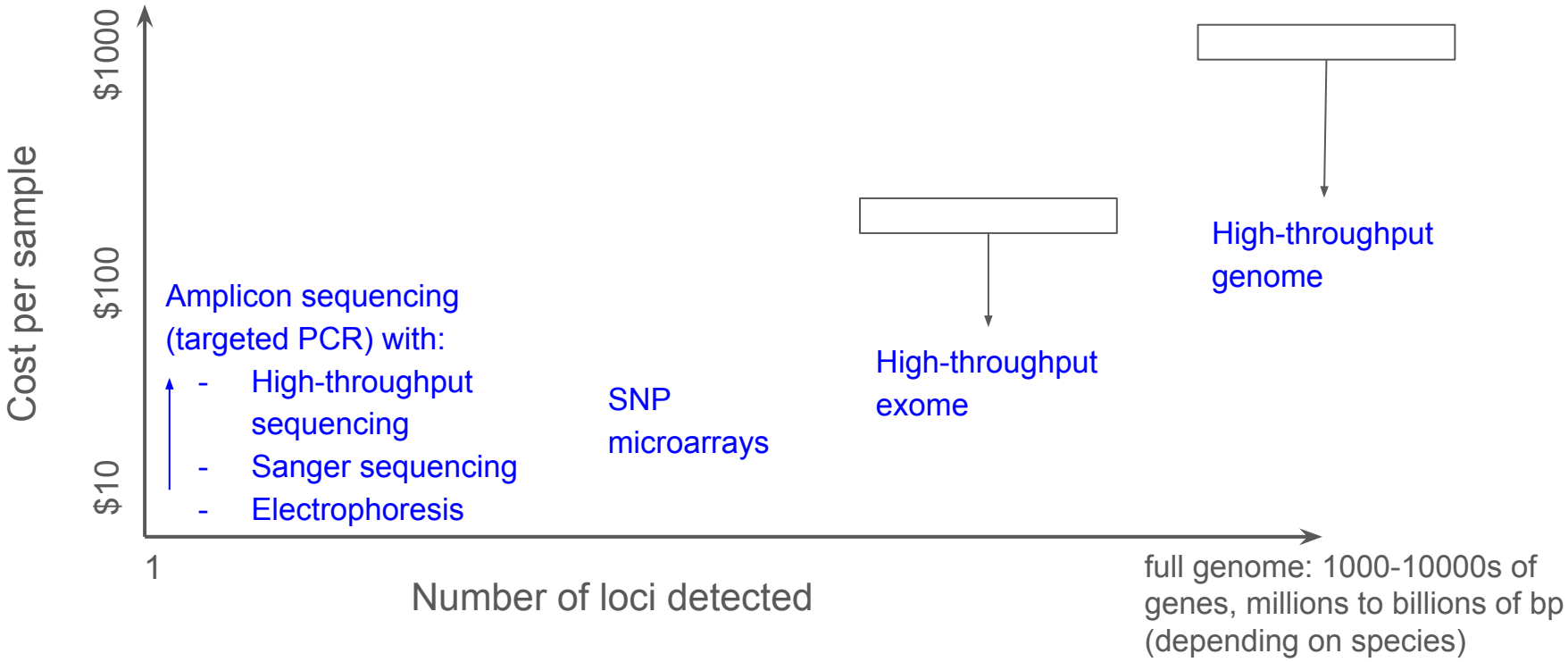
Sanger sequencing vs. high-throughput sequencing

- Sanger
  - Only sequence one locus at a time
  - Sequences are long (800-1000bp)

- High-throughput sequencing with sequencing-by-synthesis
  - Can obtain up to 16-20 billion sequences in one sequencing run (up to 48 hours)
  - Sequences are shorter (typically ~150bp)
  - Relatively cheap to sequence a whole genome or exome



Finding genome sequence variation is more inexpensive than ever



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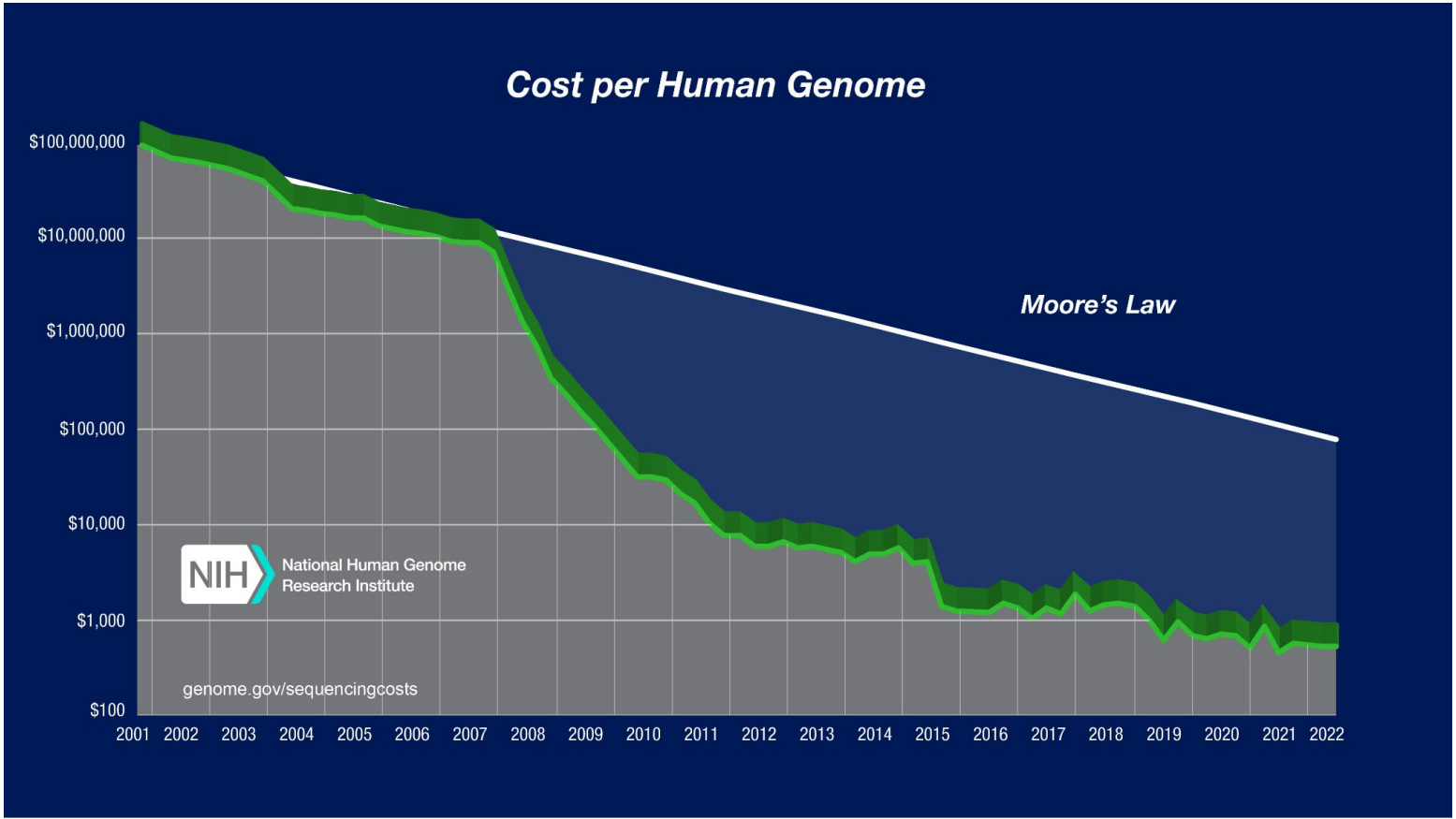
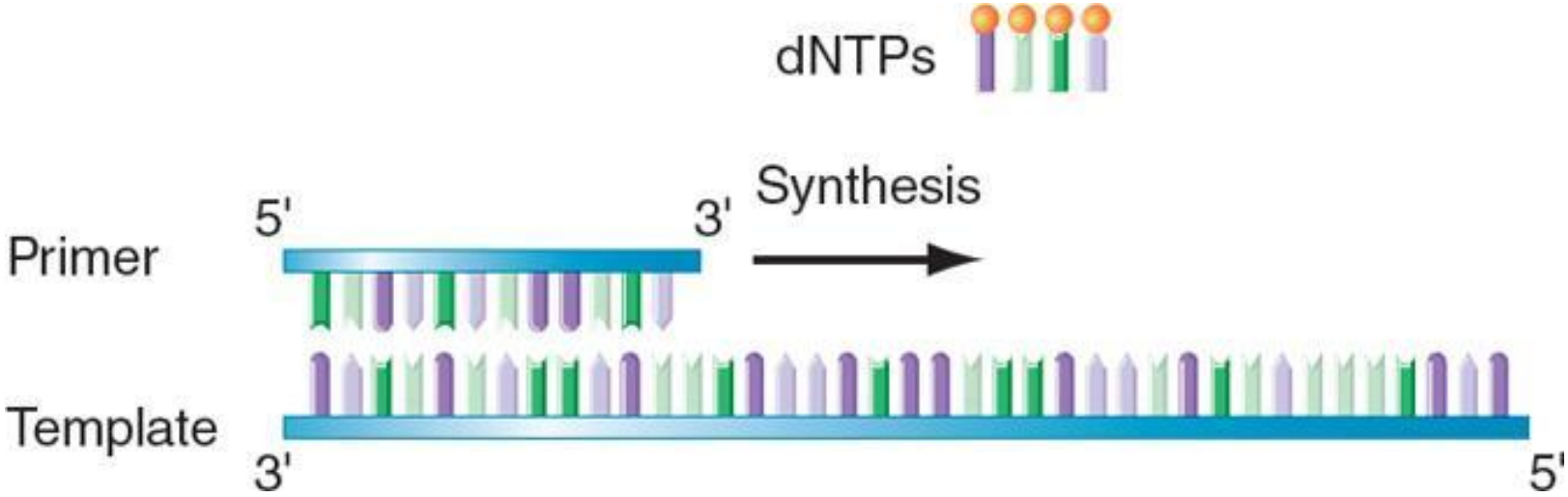
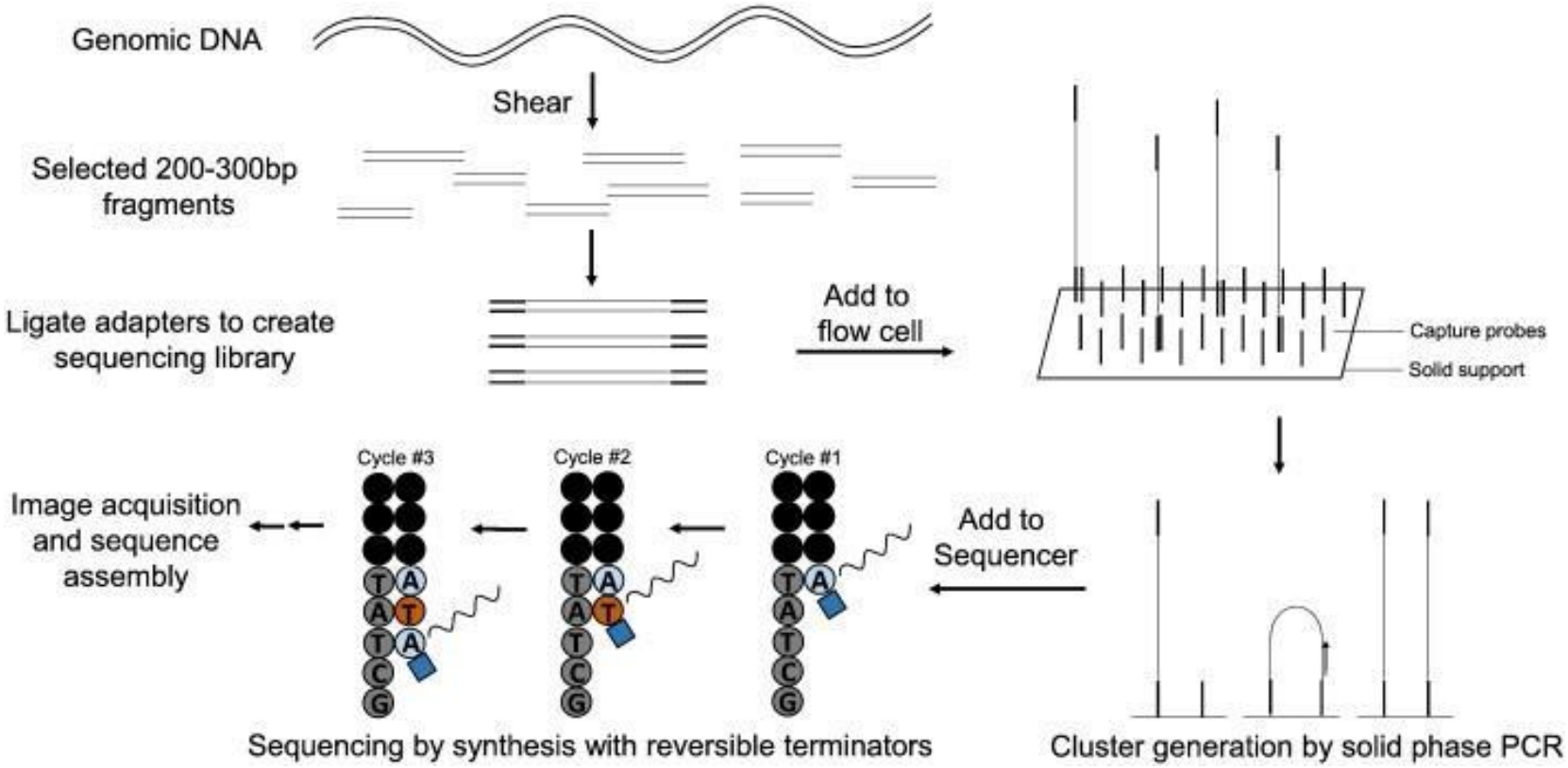


Overview of DNA synthesis: Basis of sequencing-by-synthesis

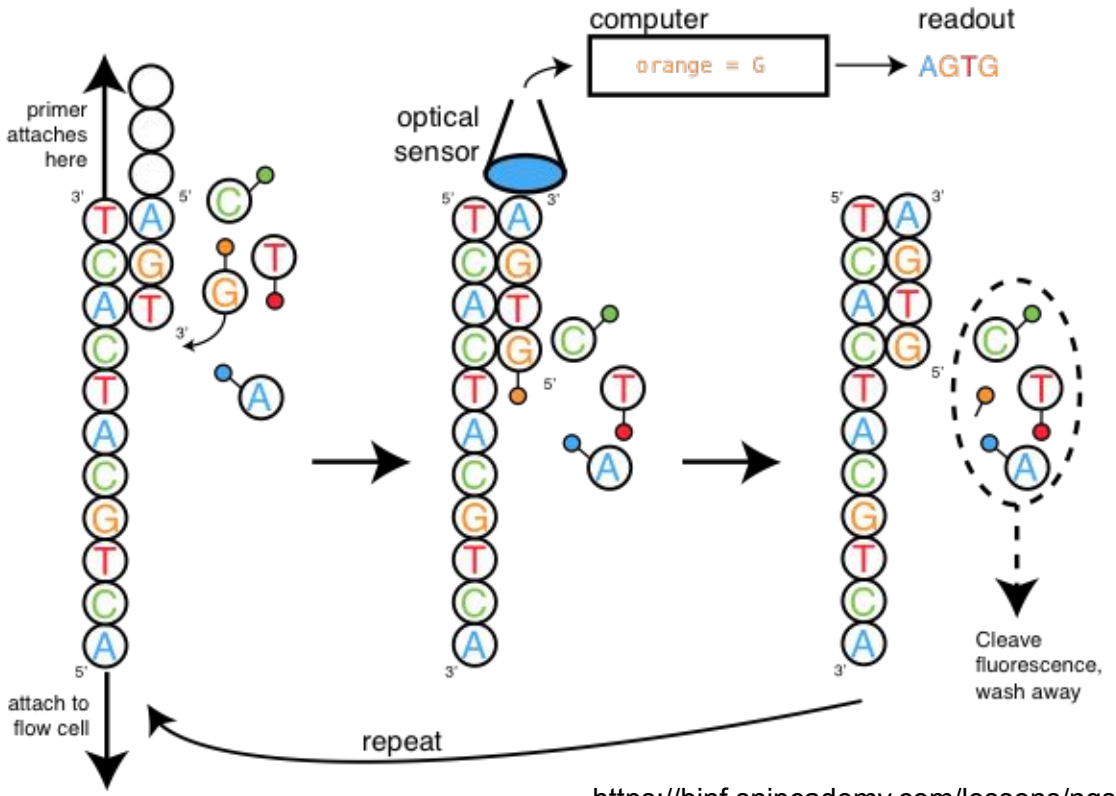


High-throughput sequencing using sequencing-by-synthesis



Biggar and Storey, Comput Struct Biotechnol J, 2014

Sequencing-by-synthesis, another view

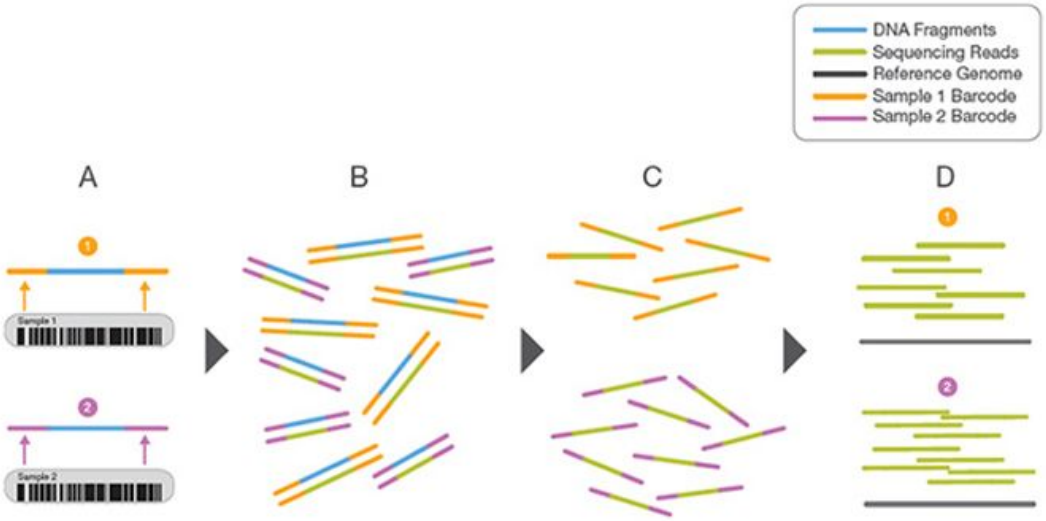


https://binf.snipcademy.com/lessons/ngs-techniques/illumina-solexa

Illumina HTS sequencing allows for sample multiplexing

Unique barcode index sequences (8-10 nt) are added to each sample during library prep

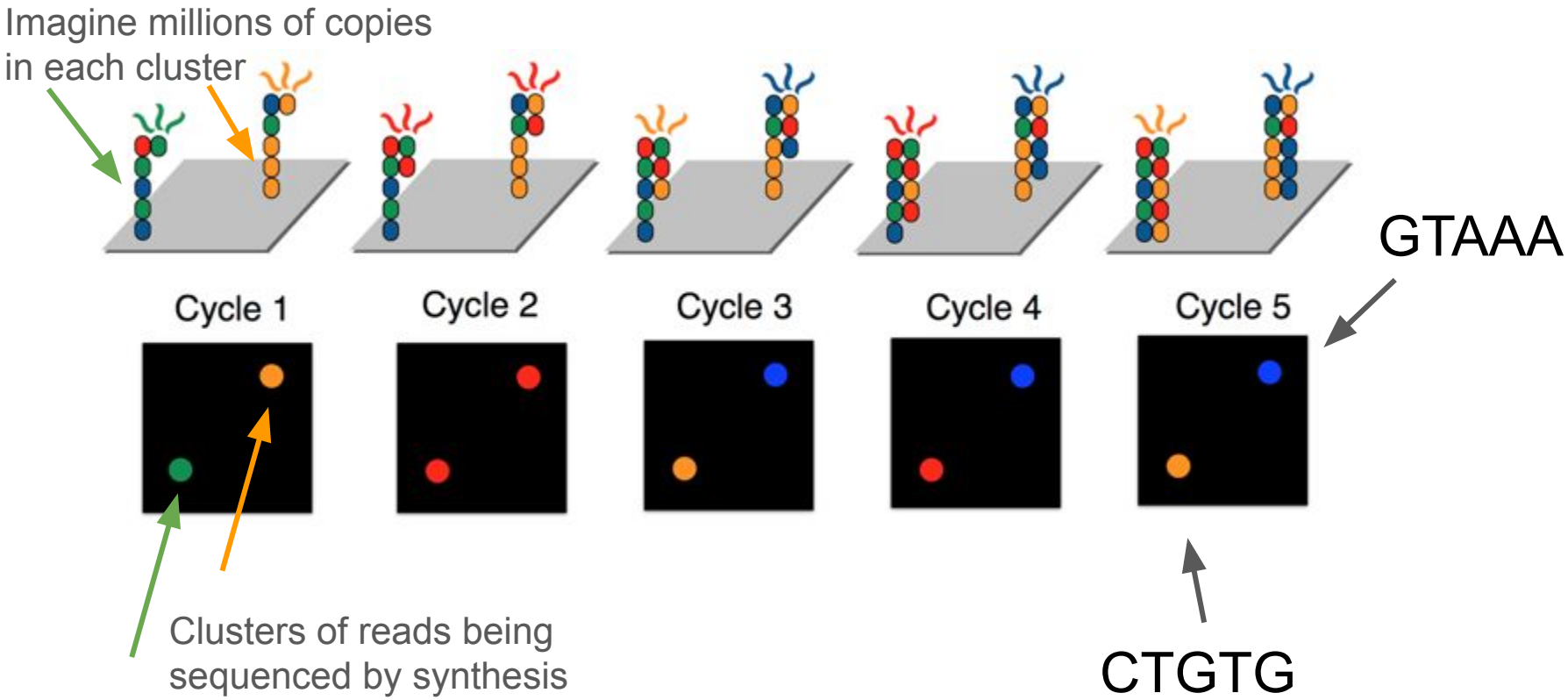
These barcodes are sequenced and used to separate sample reads bioinformatically



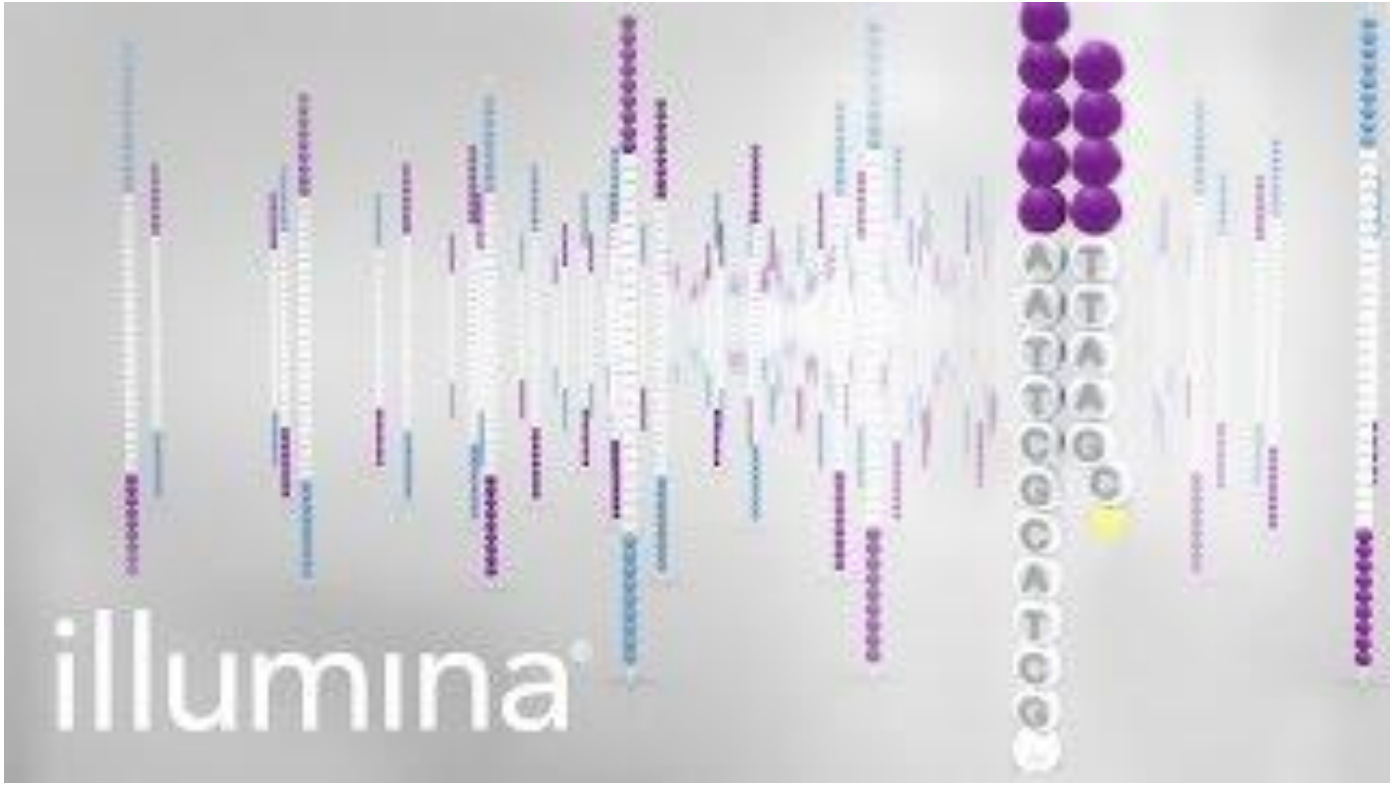
A. Two representative DNA fragments from two unique samples, each attached to a specific barcode sequence that identifies the sample from which it originated.
B. Libraries for each sample are pooled and sequenced in parallel. Each new read contains both the fragment sequence and its sample-identifying barcode.
C. Barcode sequences are used to de-multiplex, or differentiate reads from each sample.
D. Each set of reads is aligned to the reference sequence.

illumina.com

Millions of DNA pieces can be sequenced simultaneously



Illumina’s five-minute video summarizing everything we just learned:



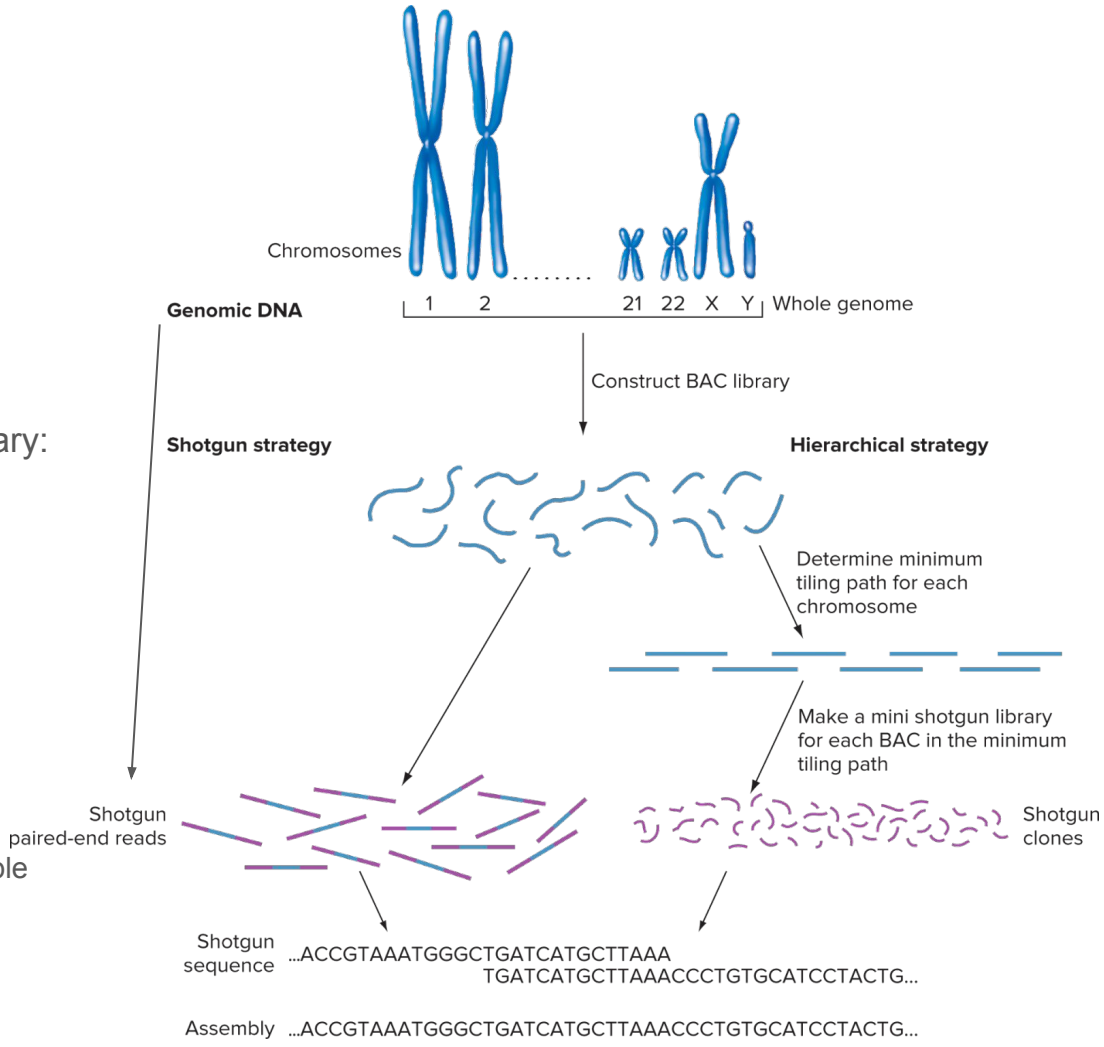
How do we sequence a genome reference?

More details in BME 130: Genomes

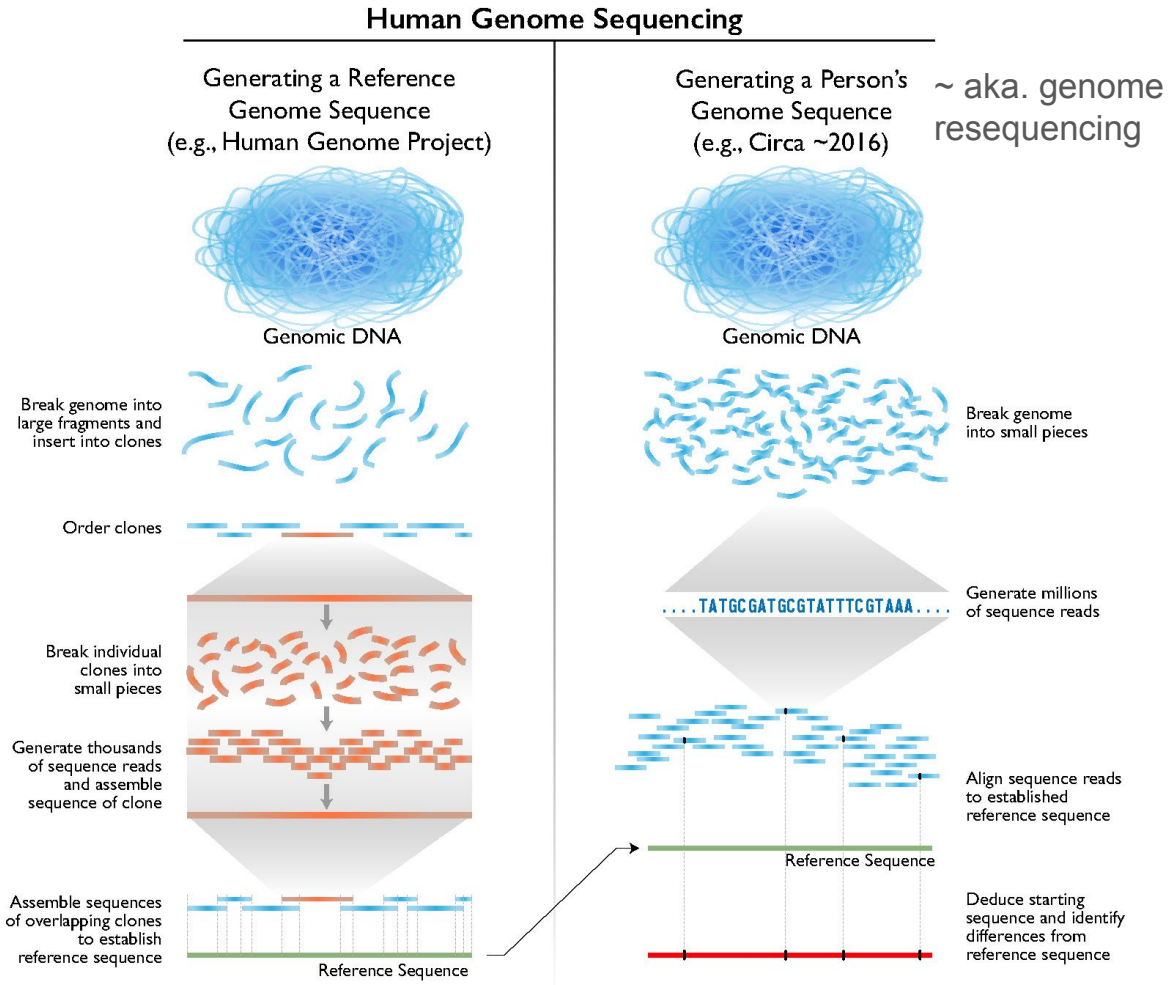
1. Direct sequencing of gDNA or

2. Bacterial artificial chromosome (BAC) library:
   - Contain 100-200 kb gDNA fragments
   - Cloned into vectors
   - Amplified in *E. coli*
   - Enable mapping of chromosomes
     - Can focus shotgun sequencing on poorly covered BACs
     - Avoid sequencing redundant BACs

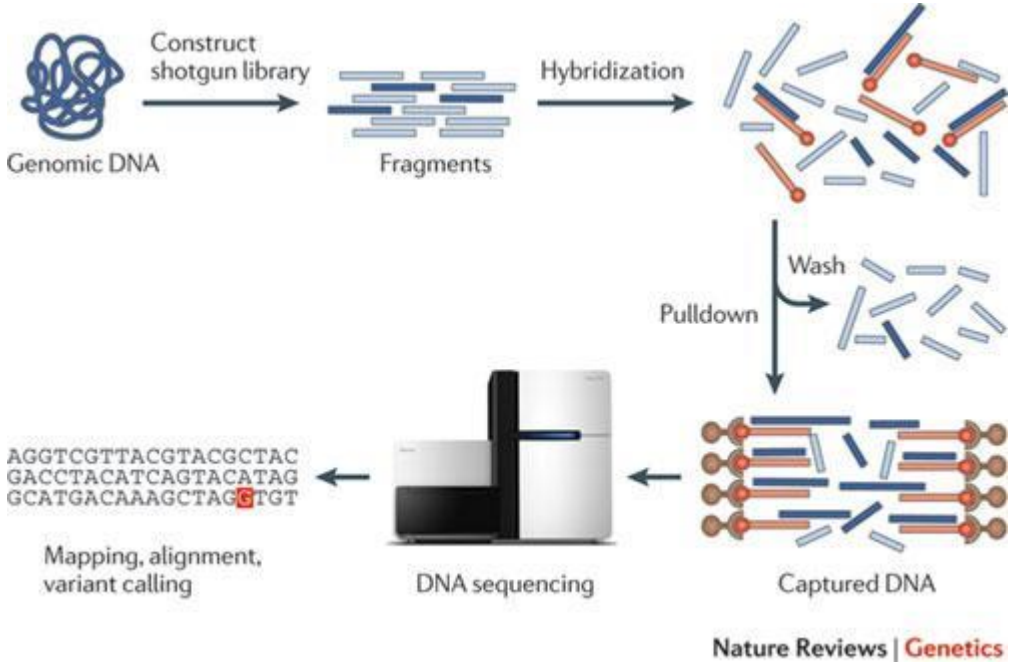
Background note: BACs are derived from stable *E. coli* F (fertility)-plasmids
- Large inserts
- Low copy number
- Faithful segregation mechanism



How do we sequence a genome reference versus many (personal) genomes?



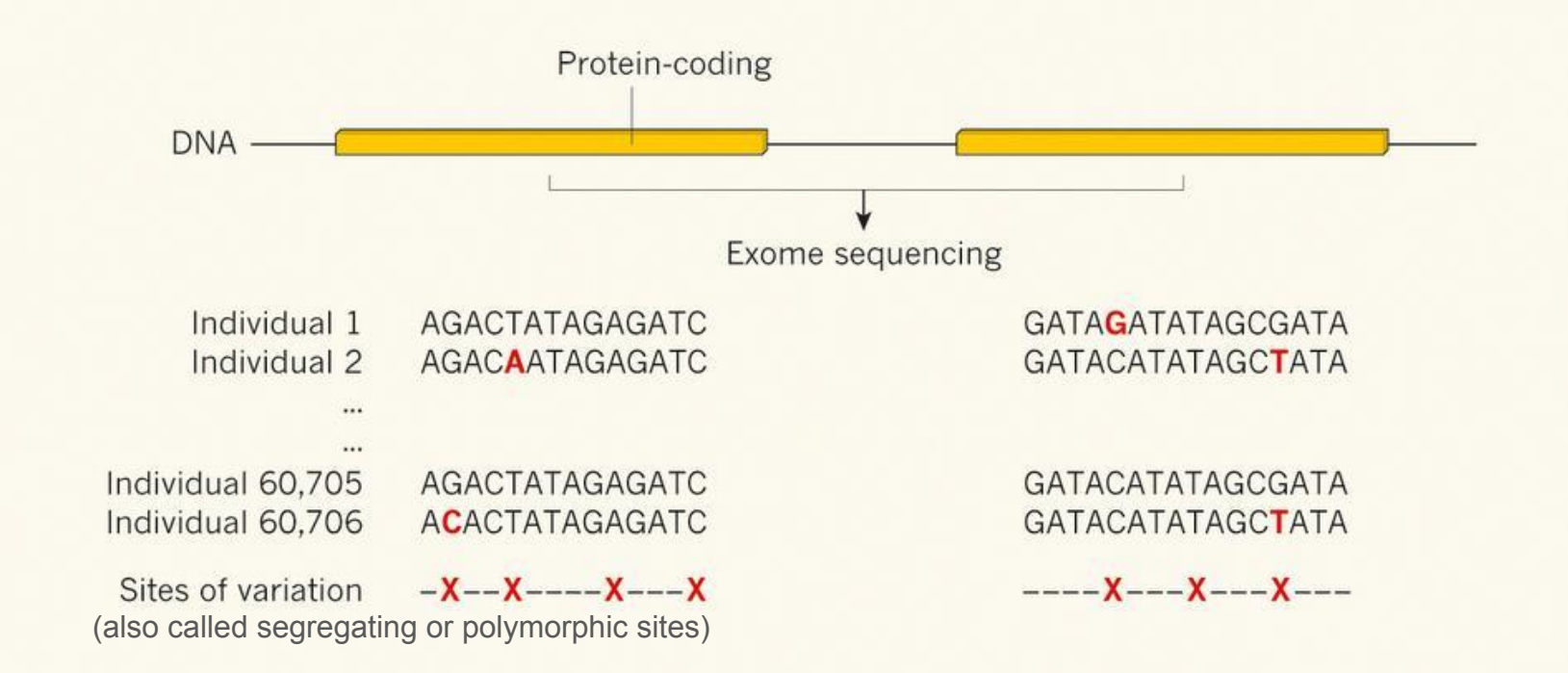
Exome sequencing to identify genetic variants in protein-coding exons



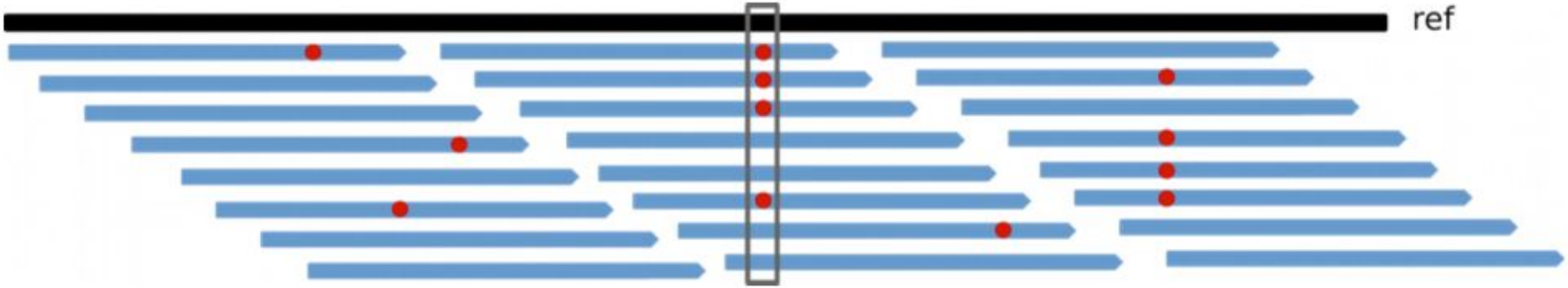
Capture probes complement exon sequence

Note: RNAseq can produce a similar dataset without the need for probes (these are sometimes called “expressed sequence tags” = EST).

Analysis of thousands of exomes can identify many sites of variation in the human genome



Identify sequence variants from high-throughput sequencing (HTS) data



Black bar: reference sequence

Blue bars: read sequence aligned to the reference

Red dot: sequence difference from reference

HTS reads are mapped (aligned) to the reference genome

```
REFERENCE:  GTACGAGCGCAAGTTGA

READ1:  GCGGAAGTTGA
```

What type of variant was sequenced?

HTS reads are mapped (aligned) to the reference genome

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REFERENCE:  GTACGAGCGCAAGTTGA

     READ1:  GTAC-------AAGTTGA
```

What type of variant was sequenced?

Issues with identifying variants from HTS

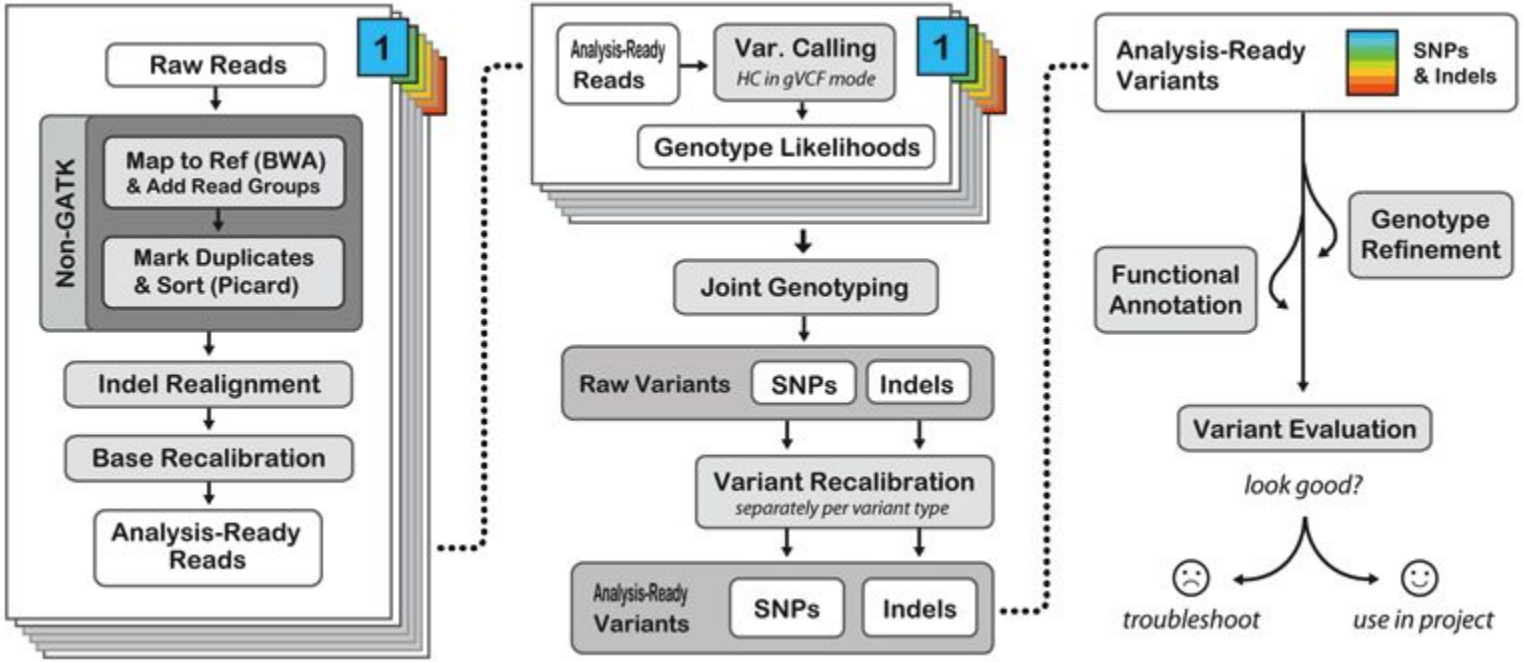
- PCR artifacts
  - Mismatches due to errors in early PCR rounds
  - PCR duplicates (can be removed bioinformatically)

- Sequencing errors
  - Erroneous mismatches either for physical reasons or properties of sequenced DNA
    - ~1% in Illumina sequencing

- Read mapping errors
  - Often happens around repeats, or other low complexity regions, and indels

Detecting variants from sequencing data with the Genome Analysis Toolkit (GATK)

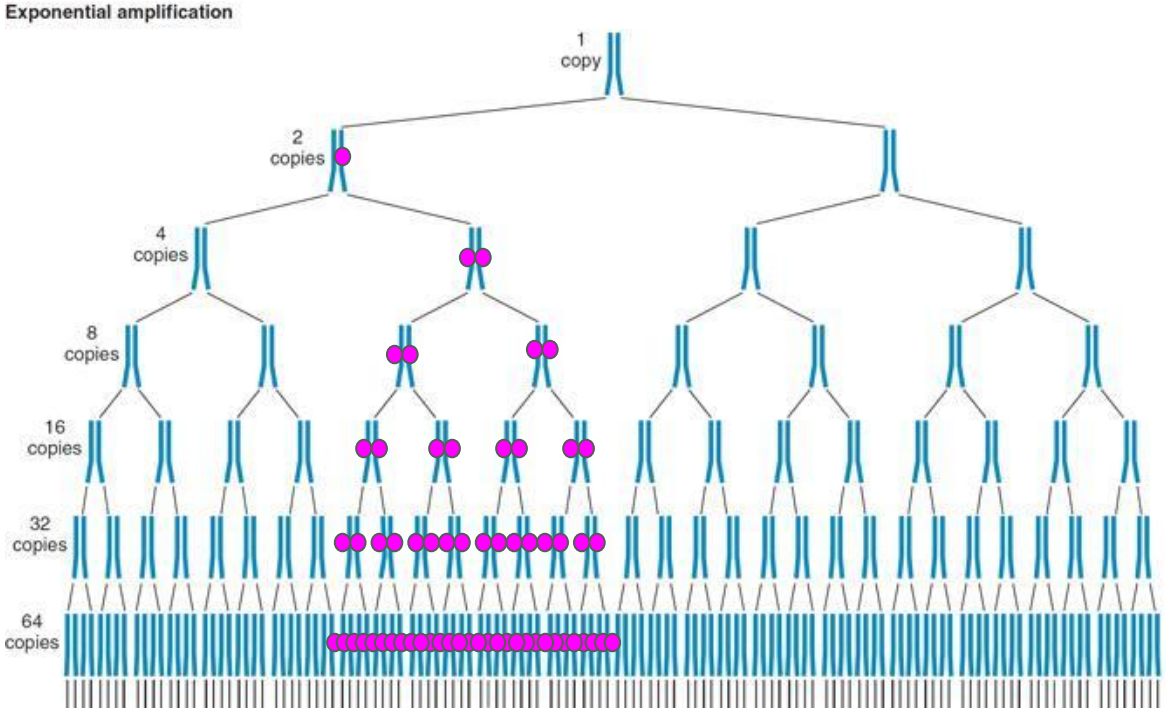
* best-practice workflow for HTS data



The Broad Institute and MIT

1. Mark duplicates

Imagine an error occurs early in PCR amplifying your library…

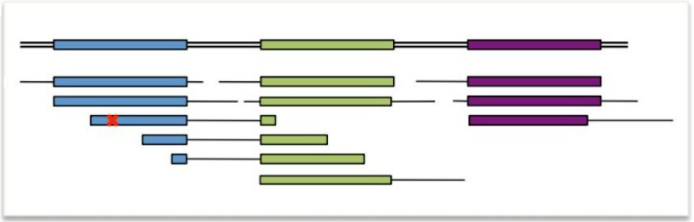


1. Mark duplicates

✖ = sequencing error propagated in duplicates

FP variant call (bad)

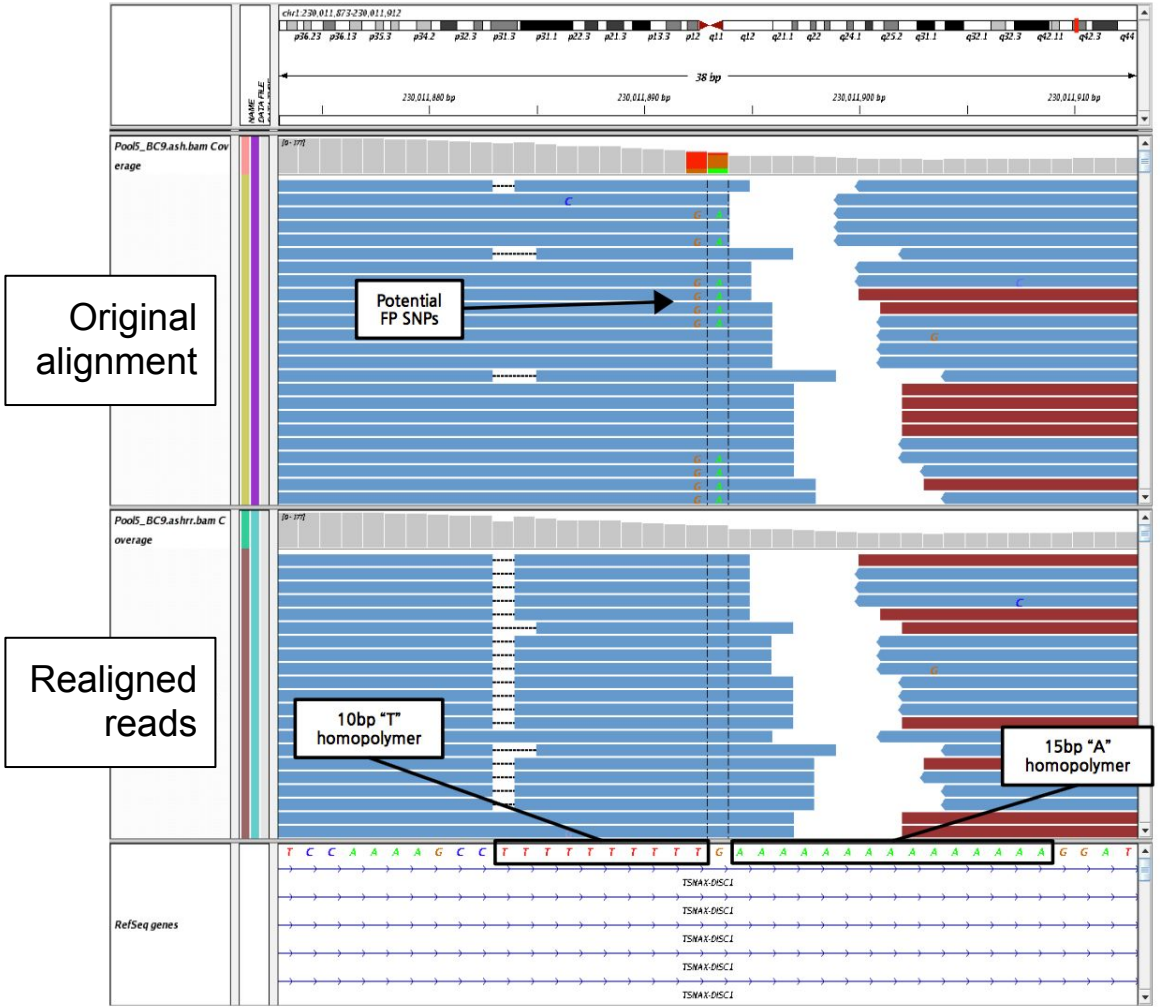
After marking duplicates, the GATK will only see :



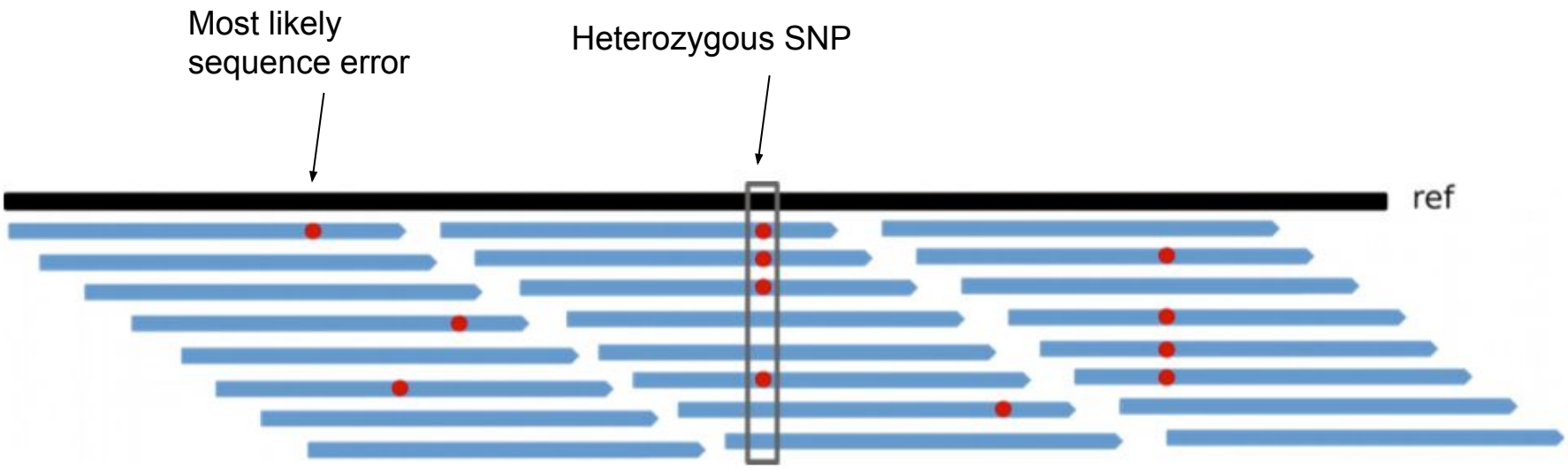
… and thus be more likely to make the right call

2. Local realignment around indels

- Reads near indels are mapped with mismatches
- Realignment can identify the most consistent placement for these reads
- Minimizes mismatches with the reference sequence
- Refines location of indels



3. Variant calling from sequencing



Black bar: reference sequence

Blue bars: read sequence aligned to the reference

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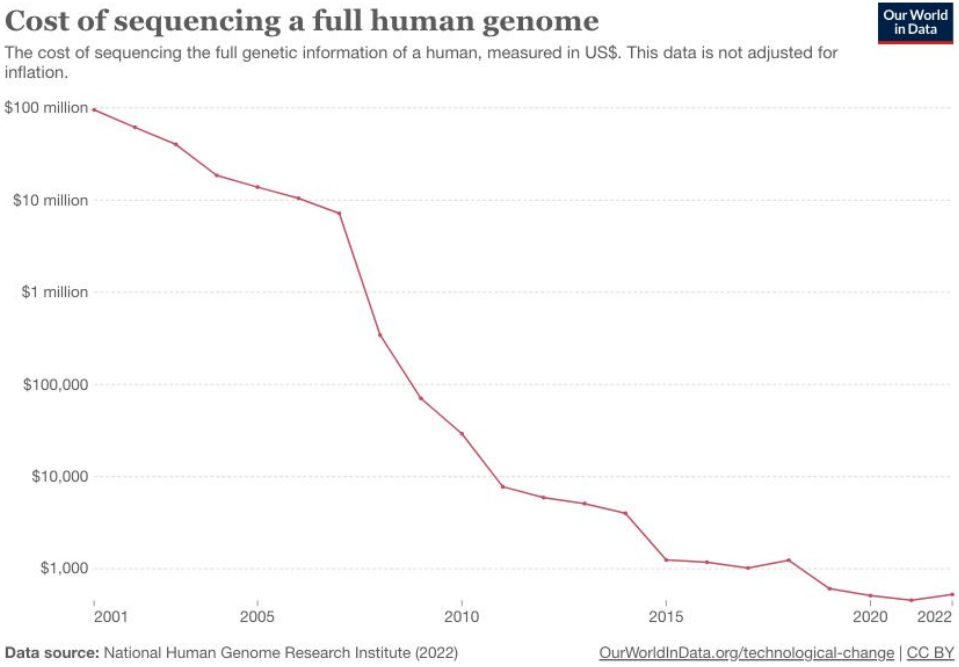
Sequencing versus genotyping

- You can use sequencing as a genotyping approach
- Genotyping is used when you already know what variant(s) to look for
- Genotyping is typically cheaper (but this is less true today)



How do we find additional (new) genome sequence variation in the human population?

- Targeted PCR followed by sequencing or gel electrophoresis
  - Sanger or gel electrophoresis - $
  - high-throughput sequencing - $$

Great for finding new alleles at known loci

- High-throughput exome ($$$) or genome sequencing($$$$)

Great for finding new structural variation and long-range haplotypes (association of alleles)

How do we decide which method to use to genotype a sample?

- Targeted PCR followed by sequencing or gel electrophoresis
  - Sanger - $
  - high-throughput sequencing - $$

When would you prefer a targeted PCR method over the other methods?

- SNP microarrays ($)

When would you prefer a SNP microarray method over the other methods?

- High-throughput exome ($$$) or genome sequencing($$$$)

When would you prefer exome or genome sequencing over other methods?

<u>Things to consider:</u>

Known variants?

Nature of variant?
- SNPs,
- indels,
- microsatellites,
- long repeats, etc.

Unknown variants?

*Possible answers in slide notes. There may be additional correct answers not listed.

How do we decide which method to use to genotype a sample?

- Targeted PCR followed by sequencing or gel electrophoresis
  - Sanger - $
  - high-throughput sequencing - $$

- SNP microarrays ($)

- High-throughput exome ($$$) or genome sequencing($$$$)

Match the sample with the method.

An individual affected with:

1. A heritable disease of unknown genetic basis
2. A known mendelian disease caused by a microsatellite expansion
3. A disease with a discrete number of possible causative SNPs

How do we decide which method to use to genotype a sample?

2. ● Targeted PCR followed by sequencing or gel electrophoresis
   - Sanger - $
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3. ● SNP microarrays

1. ● High-throughput exome ($$$) or genome sequencing($$$$)

Match the sample with the method.

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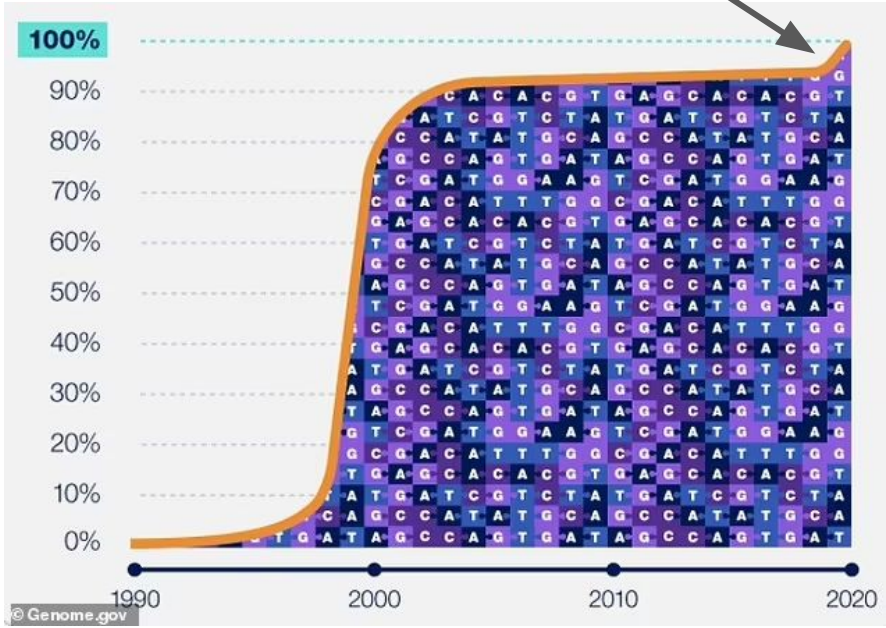
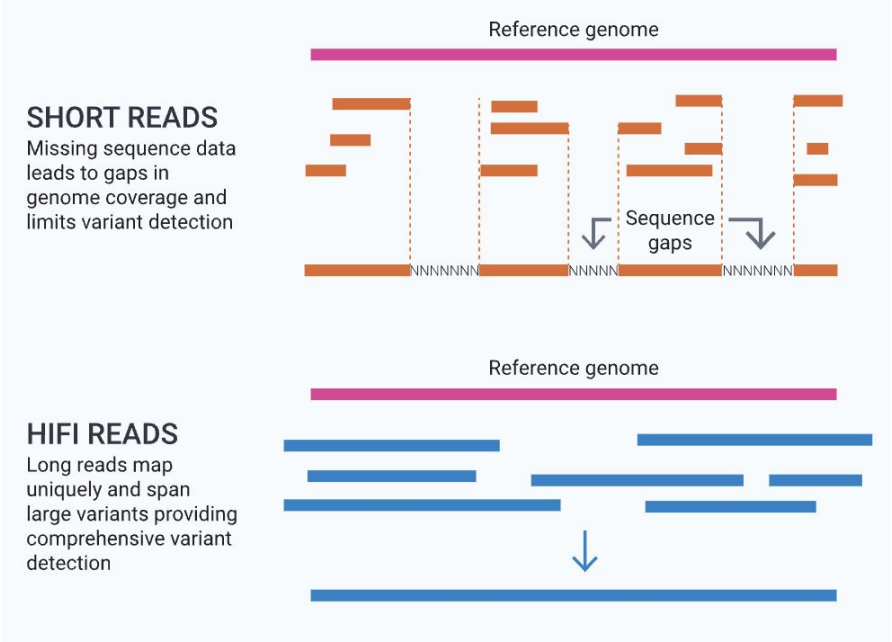
Sanger and high throughput Illumina short read sequencing are inadequate to assemble complete human genomes

Why was the last bit of sequence so hard to assemble?





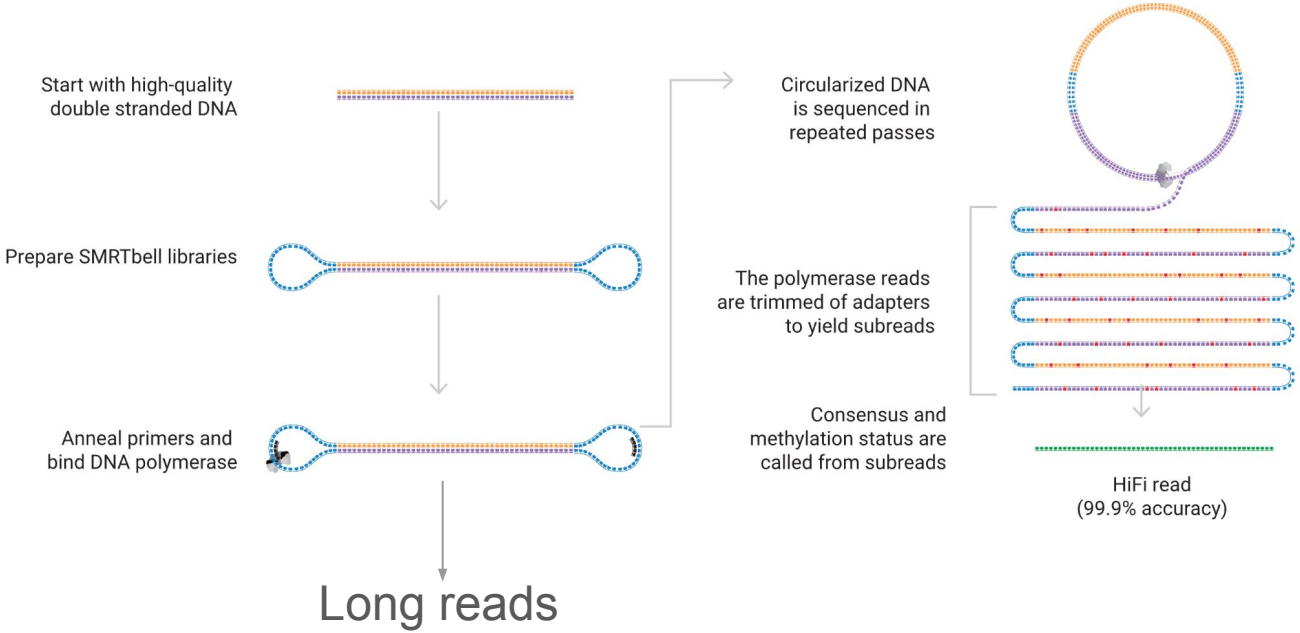
PacBio sequencing leverages DNA sequencing by synthesis of single molecules

1. Genomic DNA extraction
2. Library preparation

   Two types:

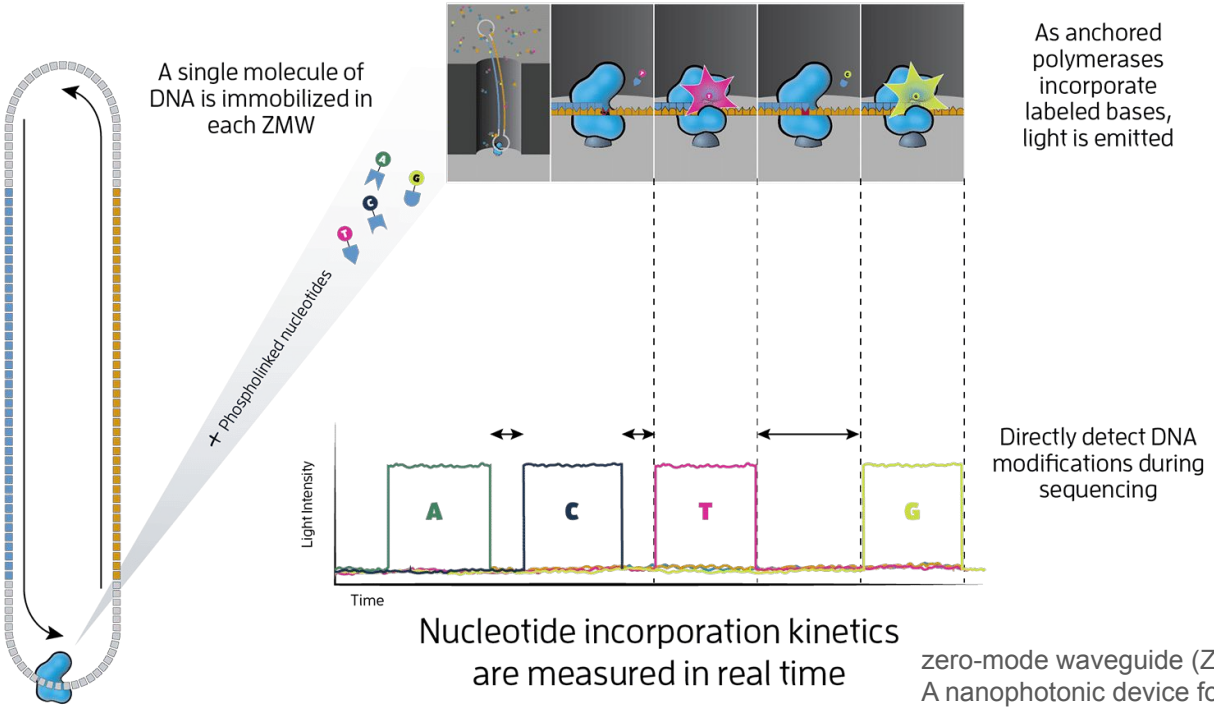
   i. Medium-accuracy (90%) 25-175 kb **long** reads

   ii. Highly accurate (99.9%) 10-20 kb **HiFi** reads



PacBio sequencing leverages DNA sequencing by synthesis of single molecules

1. Genomic DNA extraction
2. Library preparation
3. Single molecule sequencing on a PacBio sequencer platform
    a. Long reads: sequence 1x
    b. HiFi reads: sequence many times for consensus



zero-mode waveguide (ZMW): A nanophotonic device for confining light to a small observation volume

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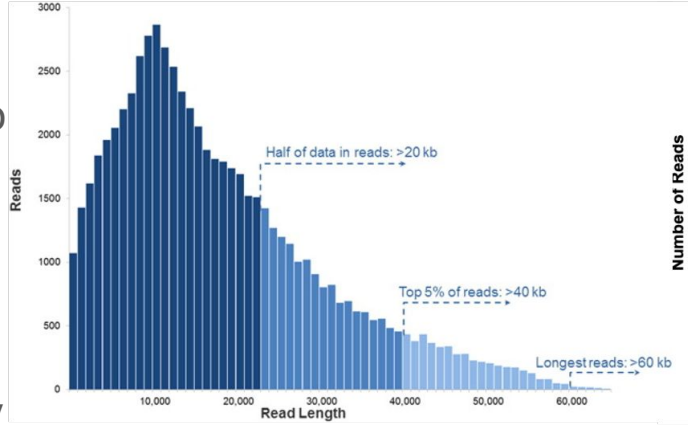
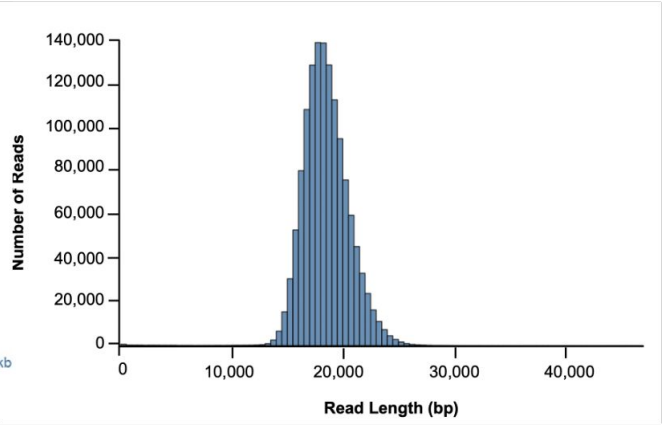


Continuous Long Reads (CLR)



HiFi Reads (CCS)

UC Davis Bioinformatic Core

How long are "long reads"?
How does Nanopore compare to PacBio?

Oxford Nanopore sequencing was invented at UCSC

Home / 2023 / September / UCSC's David Deamer and Mark Akeson honored for invention of nanopore sequencing

UCSC's David Deamer and Mark Akeson honored for invention of nanopore sequencing

September 27, 2023
By Emily Cerf

SHARE THIS STORY:

Today, two UC Santa Cruz researchers were honored at the Library of Congress for the invention of nanopore sequencing, which became a new and revolutionary method to read DNA and RNA.

David Deamer and Mark Akeson, both emeritus professors of biomolecular engineering at the Baskin School of Engineering, received the American Association for the Advancement of Science's (AAAS) Golden Goose Award for the invention. Their colleague and friend Daniel Branton, a Havard biologist and co-inventor of the technology, was also honored.

The Golden Goose award is given to scientists whose federally-funded research was seemingly

UCSC's David Deamer and Mark Akeson won the Golden Goose award for the invention of nanopore sequencing, a transformational technology for reading DNA and RNA. Here, Deamer holds the MinION, a portable device for nanopore sequencing. (photo by Nick Gonzales)

Oxford Nanopore sequencing was invented at UCSC

First paper published in 2001:

Article | Published: March 2001

Rapid discrimination among individual DNA hairpin molecules at single-nucleotide resolution using an ion channel

Wenonah Vercoutere, Stephen Winters-Hilt, Hugh Olsen, David Deamer, David Haussler & Mark Akeson

Nature Biotechnology **19**, 248–252 (2001) | Cite this article

Oxford Nanopore licenses technology from UCSC in 2008:

Oxford Nanopore announces Advisory Board, UCSC agreement

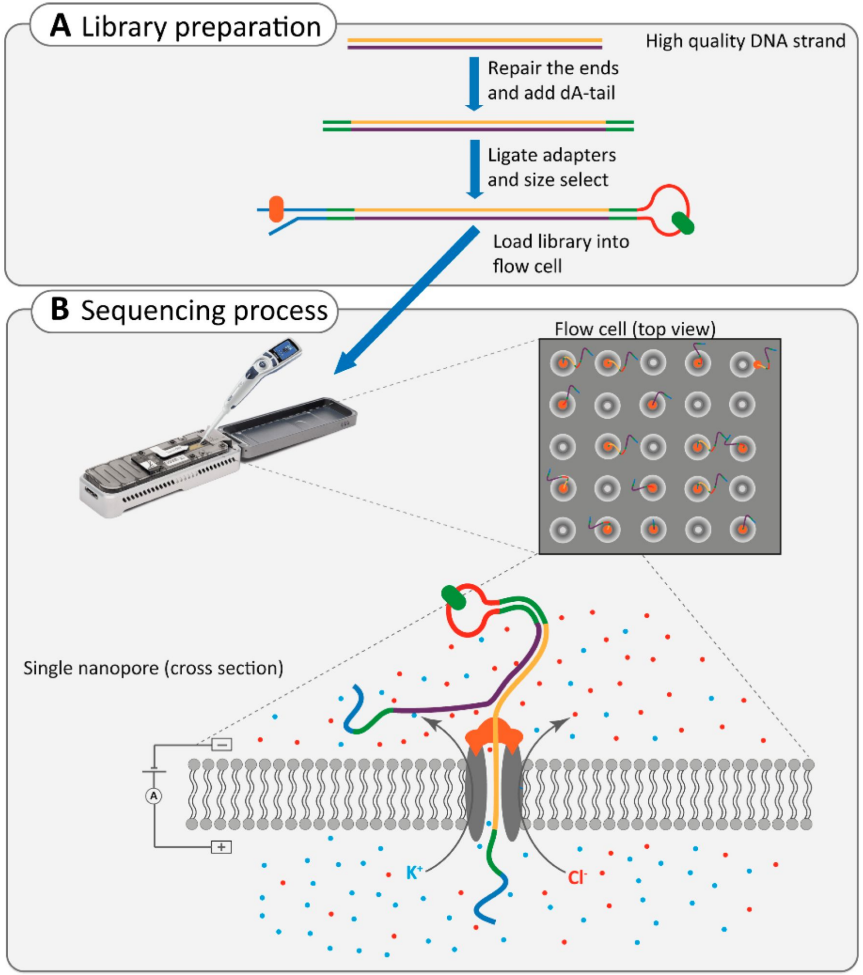
Tue 19th August 2008

Oxford Nanopore Technologies Announces License Agreement with University of California, Santa Cruz, and formation of Technology Advisory Board

August 19th 2008, Oxford, UK. Oxford Nanopore Technologies Ltd ("Oxford Nanopore") today announced the completion of an exclusive licence agreement to develop nanopore science developed at the University of California, Santa Cruz ("UCSC"), in the laboratories of Professors David Deamer and Mark Akeson.

How does Nanopore sequencing work?

1. Genomic DNA extraction
2. Library preparation (adapters for pore)
3. Library loading into flow cell (e.g., MinION)
4. DNA passes through a pore, altering the ion current through the membrane
5. These changes in the electric current are later translated to sequences.



deBruijn et al 2021 International J of Mol Sciences

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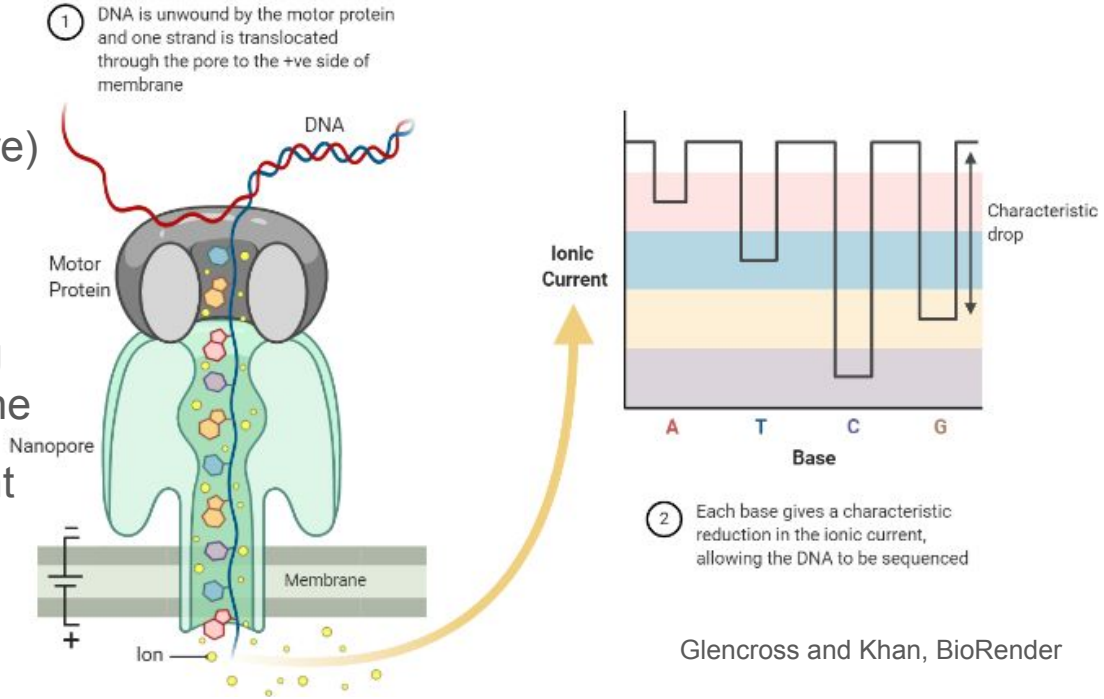
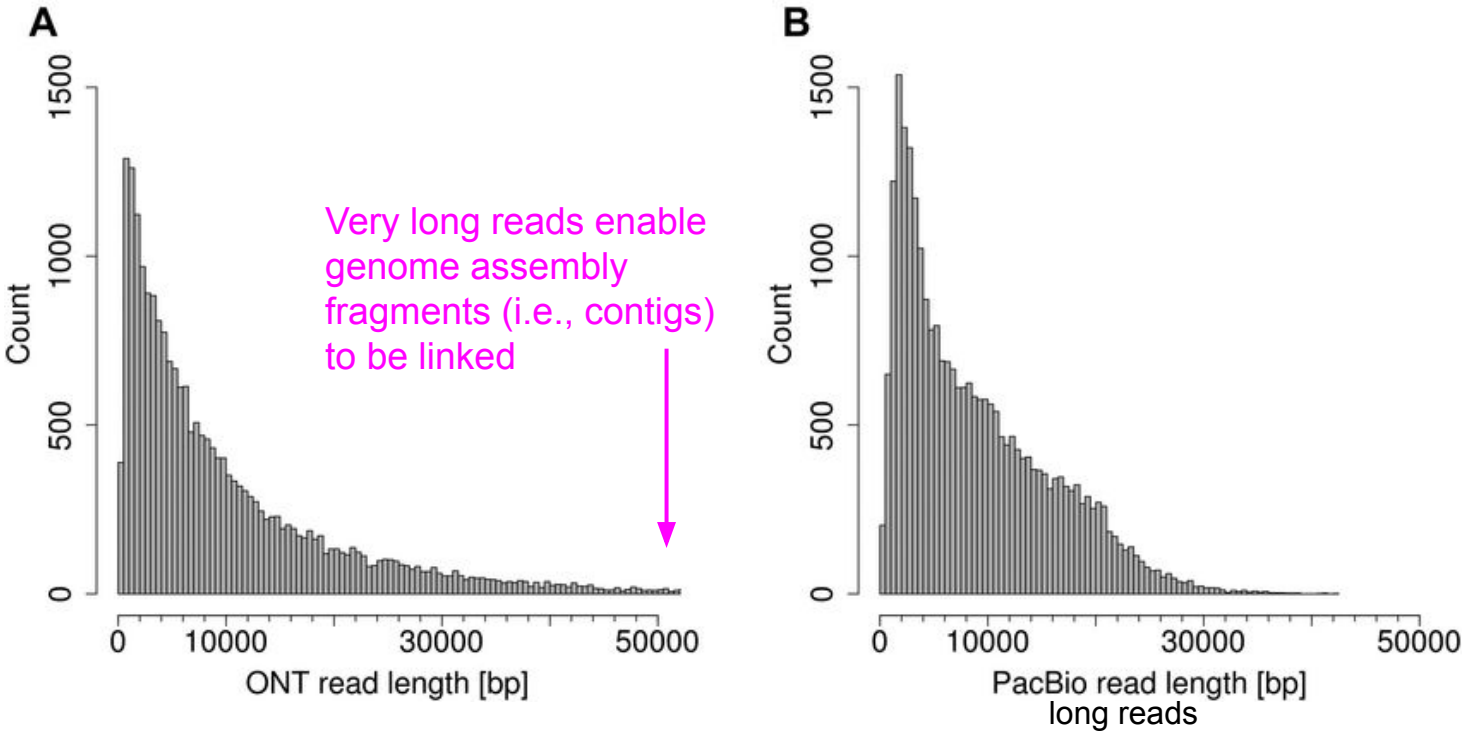


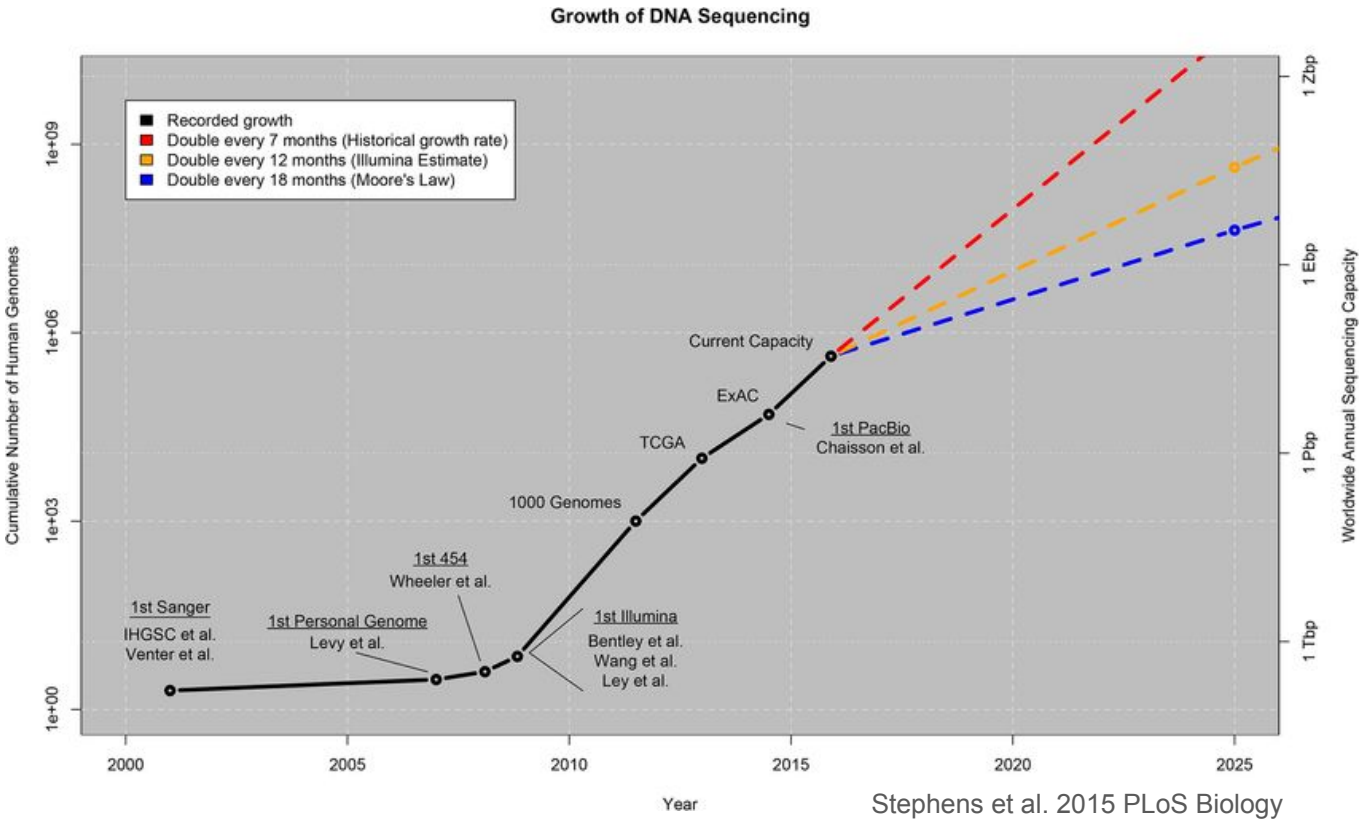
Glencross and Khan, BioRender

Nanopore is excellent at making very, very long reads



Dohm et al. 2020 Nucleic Acids Research

What do we do with all of these genomes and genetic variation?

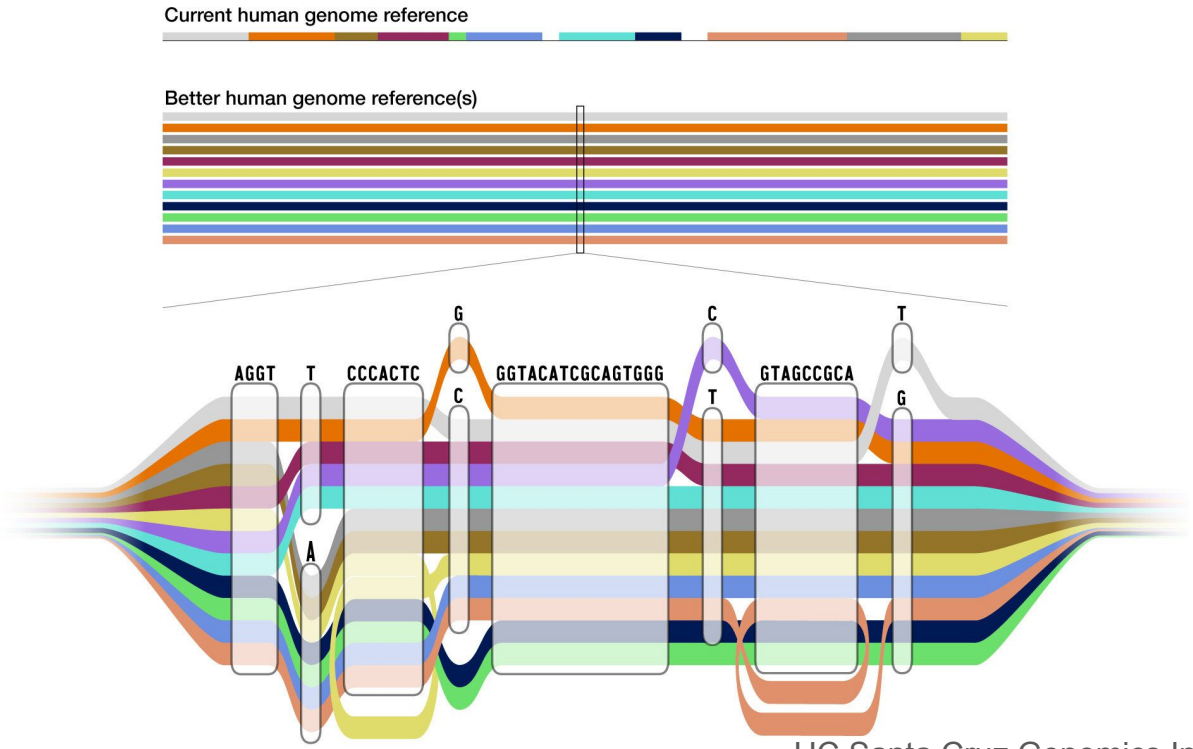


Stephens et al. 2015 PLoS Biology

Modern approaches to variation mapping: Pangenomes

Also pioneered by UCSC researchers

- Benedict Paten and the Computational Genomics Lab at UCSC

Data structure for mapping, storing, and referencing complex genetic variation

Great for incorporating short-read and long-read based variants